SYNZIP “spec sheets”

SYNZIP “spec sheets are the full data for SYNZIP pairs that have been biophysically characterized and validated in at least one in vivo assay.

Alignments are taken either from the crystal structure, or are manually curated hypothetical alignments.

The interaction table entries are as follows:
Protein microarray is the Sarray score generated from Reinke et al. 2010, ranging from 0 to 11, 0 being the strongest interaction. The two scores indicate each protein printed on the surface and probed with the interacting partner and vice versa. The Y2H scores indicate colony growth on the respective seletion media with “-” indicating no growth ranging to “+++” indicating maximum growth. These scores are manually curated from inspection of plate images and heatmap plots. MAPK scores are the fractional GFP intensity calculated as described in in the methods and range from 0 to 1, 0 being the strongest interaction. The two scores indicate a each zipper either fused to Ste5 or Msg5 unless noted otherwise.

Potential interactors are defined as: show interaction on protein microarray in at leat one direction with an Sarray score < 0.5, show an interaction in y2h in at least one direction with a pixel count > 1000. Interactions that are weak in one assay(ie pa) and strong in the other (ie y2h), are indicated as interactors for the weaker assay.

All error bars show ±1 standard deviation of measurement replicates.

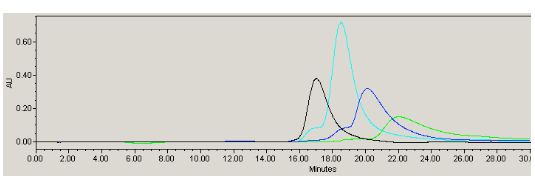
All three versions of the pENTR vector constructs, as well as the MBP-SZ-His6x constructs are available for all SYNZIPs in the following “spec sheets”

SYNZIP1:SYNZIP2

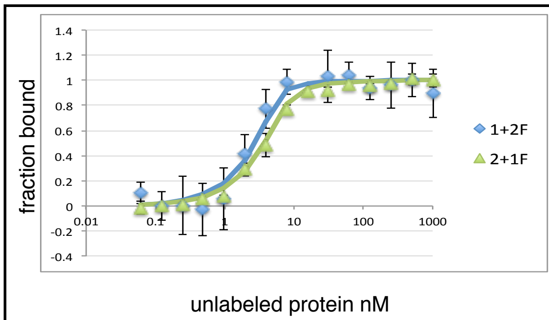
```
Alignment:
heptad position      fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdef
SZ1                  NL VAQLENE VASLENE NETLKKK NLHKKDL IAYLEKE IANLRKK IEE
SZ2                  AR NAYLRKK IARLKKD NLQLERD EQNLEKI IANLRDE IARLENE VASHEQ
-from crystal structure
```

Interaction Data

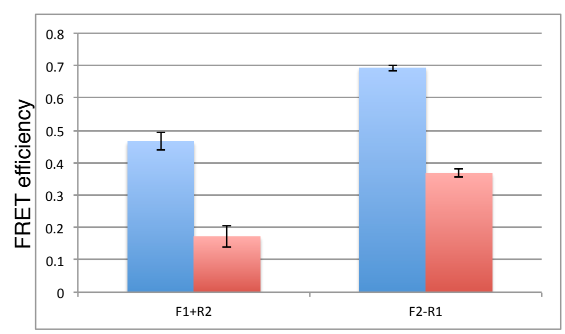
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0/0.013	++	+++	0.237/0.329	dimer	parallel	< 10 nM
	slight growth variation			SZ2 monomer possibly interacts w/ column		



Oligomerization(SEC): black- trimer control, cyan-mixture, blue-SZ1, green-SZ2



Affinity(FP): "f" denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ1: 7(y2h), 11(pa,y2h), 22(pa,y2h)

SZ2: 8(y2h), 13(pa,y2h), 14(pa,y2h), 19(pa,y2h), 20(pa,y2h)

Additional notes

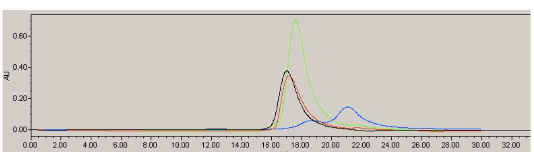
structure available: 3HE5.pdb

SYNZIP2:SYNZIP14

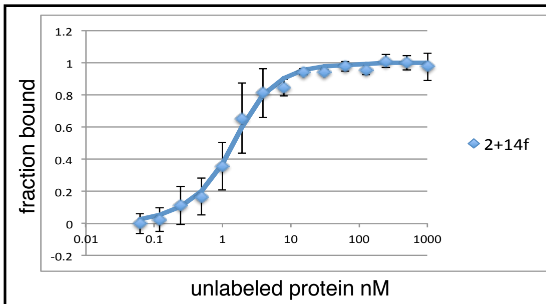
```
Alignment:
heptad position      fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdef
SZ2                  AR NAYLRKK IARLKKD NLQLERD EQNLEKI IANLRDE IARLENE VASHEQ
SZ14                 ND LDAYERE AEKLEKK NEVLRNR LAALENE LATLRQE VASMKQE LQS
hypothetical
```

Interaction Data

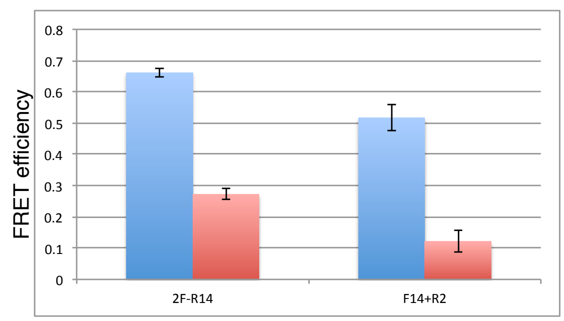
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.127/0.026	+++	+++	nd	dimer	parallel	< 10 nM
	weak one direction			2 monomer potentially interact.w/ column, 14 homodimer		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ2, red-SZ14



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ2: 1(pa,y2h), 8(y2h), 13(pa,y2h), 19(pa,y2h), 20(pa,y2h)

SZ14: 8(y2h), 12(pa,y2h), 15(pa), 17(pa), 21(pa), 22(pa,y2h)

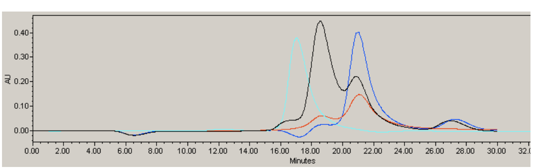
Additional notes

SYNZIP2:SYNZIP19

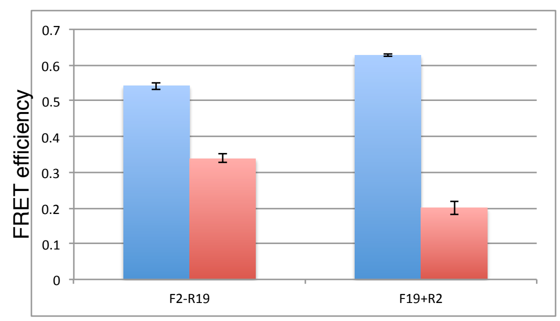
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdef
SZ2                 AR NAYLRKK IARLKKD NLQLERD EQNLEKI IANLRDE IARLENE VASHEQ
SZ19                NE LESLENK KEELKNR NEELKQK REQLKQK LAALRNK LDAYKNR L
hypothetical
```

Interaction Data

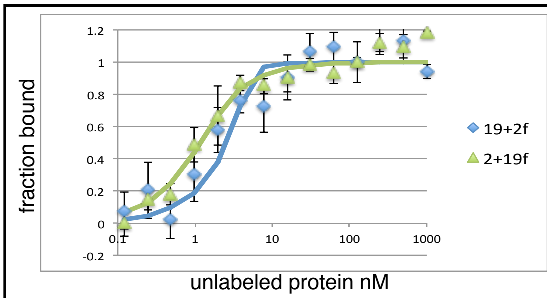
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.041/0.227	-	+++	0.340/0.468	dimer	parallel	< 10 nM
		slight growth variation		2 & 19 monomer potentially interact.w/ column		



Oligomerization(SEC): cyan- trimer control, green-mixture, blue-SZ1, red-SZ2



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor



Affinity(FP): “f” denotes labeled species

Potential interactors

SZ2: 1(pa,y2h), 8(y2h), 13(pa,y2h), 14(pa,y2h), 20(pa,y2h)

SZ19: 6(pa,y2h), 11(pa,y2h), 12(pa,y2h), 16 (pa,y2h), 18(pa,y2h), 21(pa,y2h), 22 (pa,y2h)

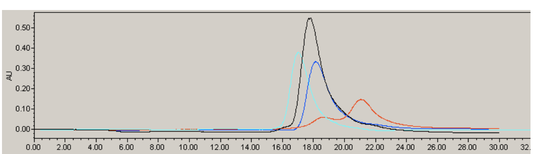
Additional notes

SYNZIP2:SYNZIP20

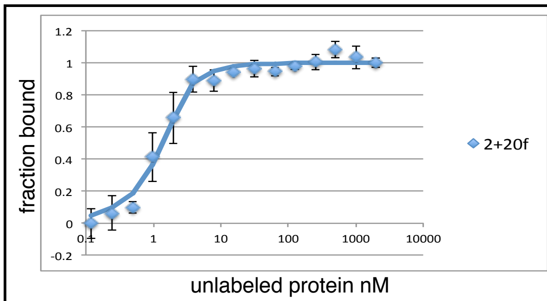
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdef
SZ2                 AR NAYLRKK IARLKKD NLQLERD EQNLEKI IANLRDE IARLENE VASHEQ
SZ20                ST VEELLRA IQELEKR NAELKNR KEELKNL VAHLRQE LAAHKYE
hypothetical
```

Interaction Data

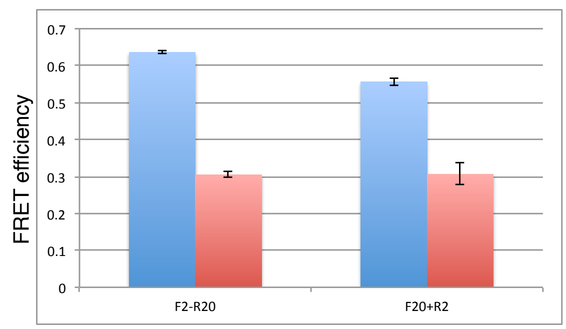
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0/0	-	+++	0.315	dimer	parallel	< 10 nM
			Ste5:SZ20-Msg5:SZ2 tested	2 monomer potentially interact.w/ column, 20 homodimer		



Oligomerization(SEC): cyan- trimer control, black-mixture, blue-SZ20, red-SZ2



Affinity(FP): "f" denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ2: 1 (pa,y2h), 8 (y2h), 13 (pa,y2h), 14 (pa,y2h), 19 (pa,y2h)

SZ20: 3 (pa), 6 (pa,y2h), 11 (pa,y2h), 12 (pa,y2h), 16 (pa,y2h), 18 (pa,y2h), 21 (pa,y2h), 22 (pa,y2h)

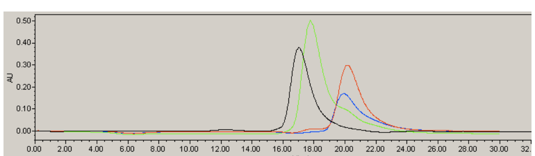
Additional notes

SYNZIP3:SYNZIP4

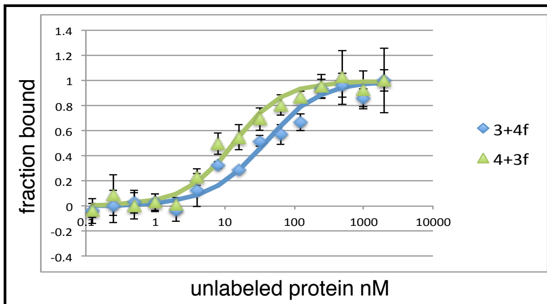
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcde
SZ3                                NE VTTLEND AAFIENE NAYLEKE IARLRKE KAALRNR LAHKK
SZ4                 QK VAELKNR VAVKLNR NEQLKNK VEELKNR NAYLKNE LATLENE VARLEND VAE
-from CD deletion assay
```

Interaction Data

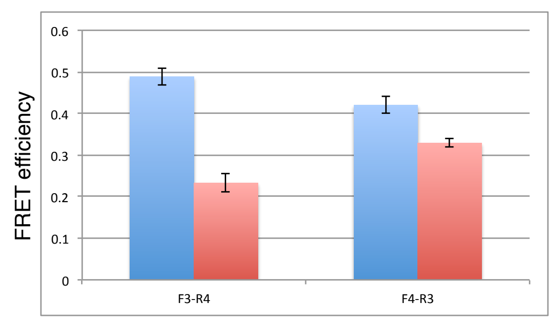
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0/0.079	+++	+++	0.219/0.366	dimer	parallel	< 30 nM
	4 autoactivation	4 autoactivation				



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ3, red-SZ4



Affinity(FP): "f" denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ3: 5(pa,y2h) 13(pa,y2h), 17(pa,y2h), 20(pa,y2h)

SZ4: 6(pa,y2h), 18(pa,y2h), 21(pa,y2h), 22(y2h)

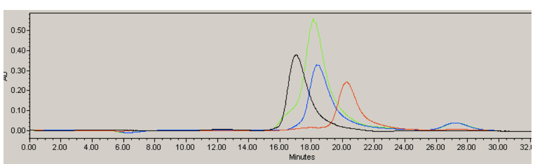
Additional notes

SYNZIP4:SYNZIP21

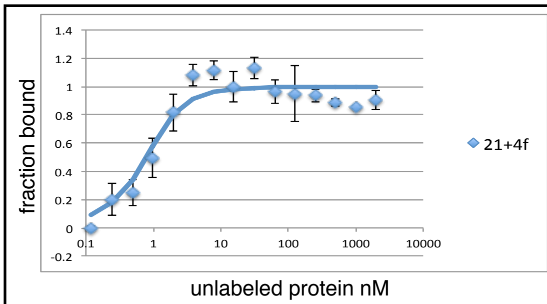
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcde
SZ4                 QK VAELKNR VAVKLNR NEQLKNK VEELKNR NAYLKNE LATLENE VARLEND VAE
SZ21                                NE VAQLEND VAVIENE NAYLEKE IARLRKE IAALRDR LAHKK
hypothetical
```

Interaction Data

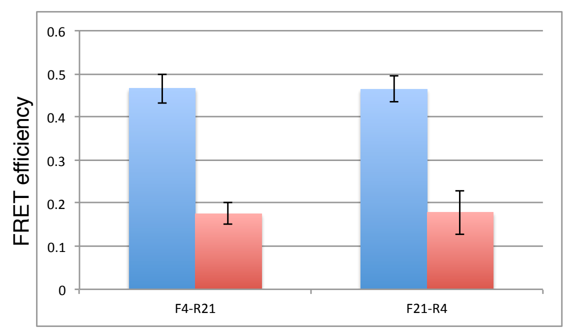
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.135/0.137	+++	+++	0.480	dimer	parallel	< 10 nM
	4 autoacti vation	4 autoactiv ation	Ste5:SZ4 Msg5:SZ21 tested	21 homodimer		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ21, red-SZ4



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ4: 3(pa,y2h), 6(pa,y2h), 18(pa,*y2h*), 22(y2h)
SZ21: 5(pa,y2h), 8(y2h), 10(pa,y2h), 11(pa,y2h), 12(y2h), 13(pa,y2h), 14(pa), 15(pa), 16(pa,y2h), 17(pa,y2h), 19(pa,y2h), 20(pa,y2h)

Additional notes

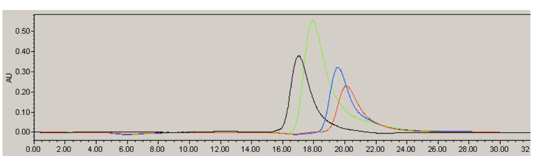
SYNZIP5:SYNZIP6

Alignment:

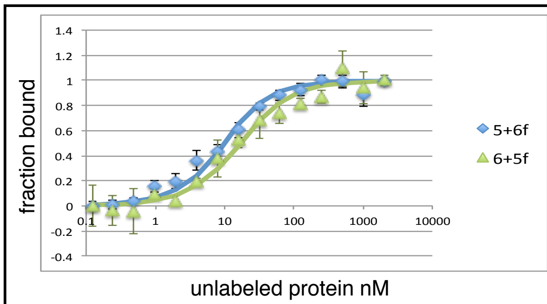
```
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg
SZ5                                NT VKELKNY IQELEER NAELKNL KEHLKFA KAELEFE LAAHKFE
SZ6                 QK VAQLKNR VAYKLKE NAKLENI VARLEND NANLEKD IANLEKD IANLERD VAR
-from crystal structure
```

Interaction Data

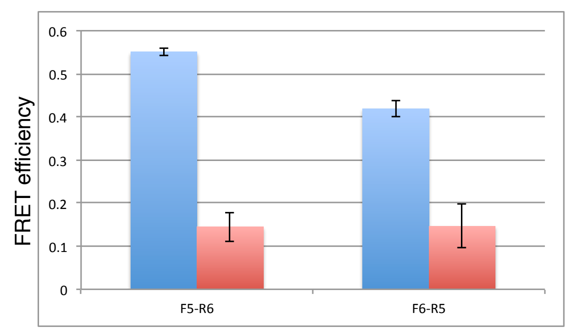
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0/0	+++	+++	0.371/0.290	dimer	parallel	< 15 nM



Oligomerization(SEC): black- trimer control, green- mixture, blue-SZ6, red-SZ5



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ5: 3 (pa,y2h), 11 (pa,y2h), 12 (pa,y2h), 16 (pa,y2h), 18 (pa,y2h), 21 (pa,y2h), 22 (pa,y2h)

SZ6: 4 (pa,y2h), 14 (y2h), 17 (pa,y2h), 19 (pa,y2h), 20 (pa,y2h)

Additional notes

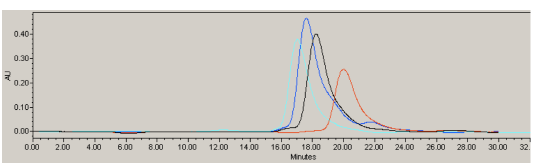
structure available: 3HE4.pdb

SYNZIP5:SYNZIP16

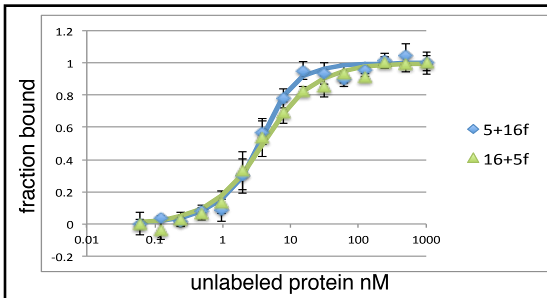
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg
SZ5                 NT VKELKNY IQELEER NAELKNL KEHLKFA KAELEFE LAAHKFE
SZ16                NI LASLENK KEELKKL NAHLLKE IENLEKE IANLEKE IAYFK
hypothetical
```

Interaction Data

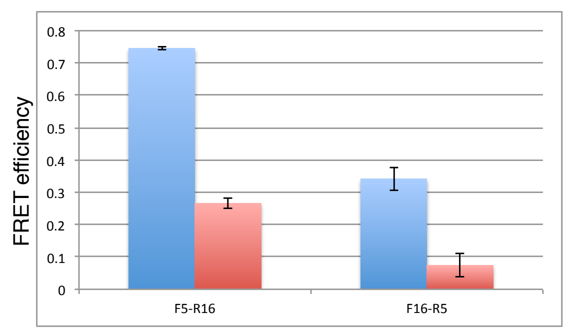
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.339/0.2	+++	+++	nd	dimer	parallel	< 10 nM



Oligomerization(SEC): cyan- trimer control, black-mixture, blue-SZ16, red-SZ5



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ5: 3[pa,y2h], 6[pa,y2h], 11[pa,y2h], 12[pa,y2h], 18[pa,y2h], 21[pa,y2h], 22[pa,y2h]

SZ16: 19[pa,y2h], 20[pa,y2h], 21[pa,y2h]

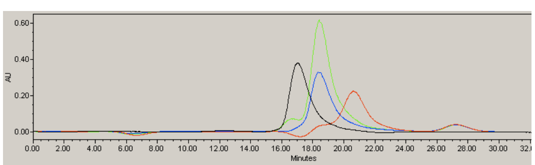
Additional notes

SYNZIP5:SYNZIP21

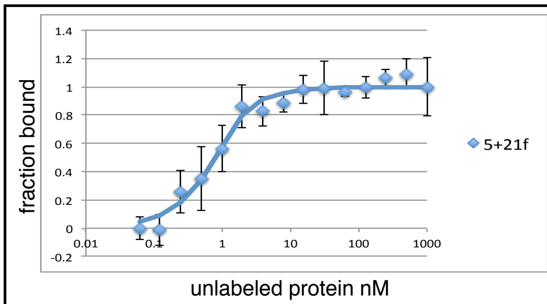
```
Alignment:
heptad position      fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg
SZ5                  NT VKELKNY IQELEER NAELKNL KEHLKFA KAELEFE LAAHKFE
SZ21                 NE VAQLEND VAVIENE NAYLEKE IARLRKE IAALRDR LAHKK
hypothetical
```

Interaction Data

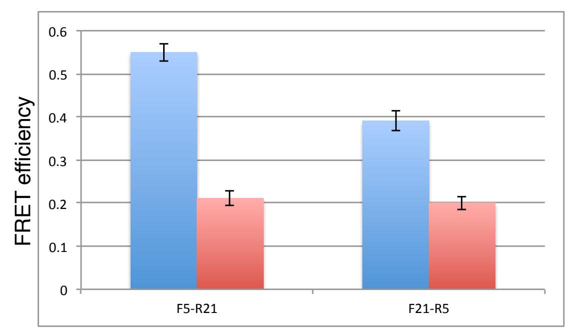
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.18/0.22	+++	+++	nd	dimer	parallel	< 10 nM
				21 homodimer		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ21, red-SZ5



Affinity(FP): "f" denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ5: 3(pa,y2h), 6(pa,y2h), 11(pa,y2h), 12(pa,y2h), 16(pa,y2h), 18(pa,y2h), 22(pa,y2h)

SZ21: 4(pa,y2h), 8(y2h), 10(pa,y2h), 11(pa,y2h), 12(y2h), 13(pa,y2h), 14(pa), 15(pa), 16(pa,y2h), 17(pa,y2h), 19(pa,y2h), 20(pa,y2h)

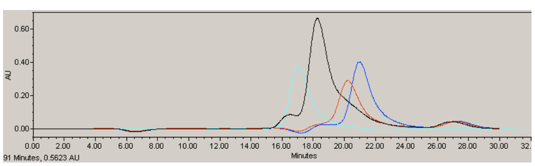
Additional notes

SYNZIP6:SYNZIP19

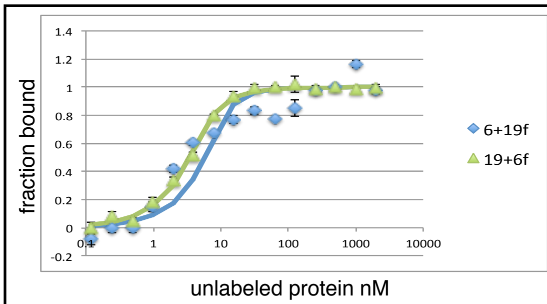
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg a
SZ6                 QK VAQLKNR VAYKLKE NAKLENI VARLEND NANLEKD IANLEKD IANLERD VAR
SZ19                                   NE LESLENK KEELKNR NEELKQK REQLKQK LAALRNK LDAYKNR L
hypothetical
```

Interaction Data

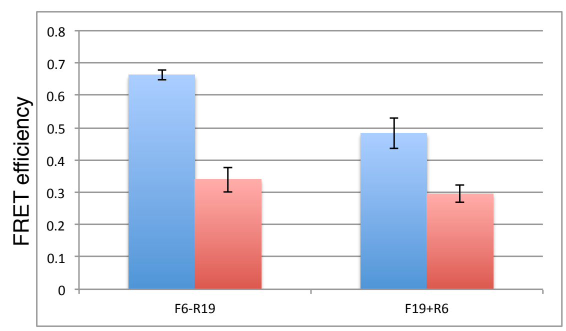
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.241/0	+	+++	nd	dimer	parallel	< 10 nM
	growth variation			19 monomer potentially interact.w/ column		



Oligomerization(SEC): cyan- trimer control, black-mixture, blue-SZ19, red-SZ6



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ6: 4(pa,y2h), 5(pa,y2h),14 (y2h), 17(pa,y2h), 20(pa,y2h)

SZ19: 2(pa,y2h), 11(pa,y2h), 12(pa,y2h), 16 (pa,y2h), 18(pa,y2h), 21(pa,y2h), 22 (pa,y2h)

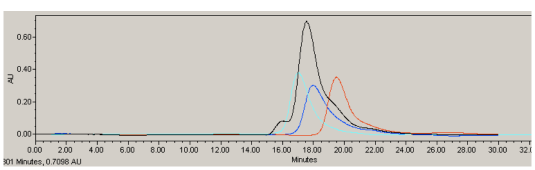
Additional notes

SYNZIP6:SYNZIP20

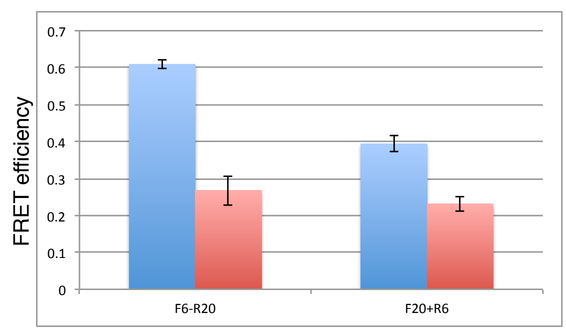
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg
SZ6                 QK VAQLKNR VAYKLKE NAKLENI VARLEND NANLEKD IANLEKD IANLERD VAR
SZ20                                ST VEELLRA IQELEKR NAELKNR KEELKNL VAHLRQE LAAHKYE
hypothetical
```

Interaction Data

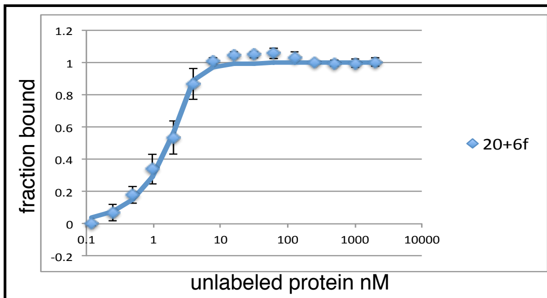
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.273/0.004	++	+++	0.582/0.685	dimer	parallel	< 10 nM
	growth variation			20 homodimer, runs as large dimer		



Oligomerization(SEC): cyan- trimer control, black-mixture, blue-SZ20, red-SZ6



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor



Affinity(FP): “f” denotes labeled species

Potential interactors

SZ6: 4 (pa,y2h), 5 (pa,y2h), 14 (y2h), 17 (pa,y2h), 19 (pa,y2h)

SZ20: 2 (pa,y2h), 3 (pa), 11 (pa,y2h), 12 (pa,y2h), 16 (pa,y2h), 18 (pa,y2h), 21 (pa,y2h), 22 (pa,y2h)

Additional notes

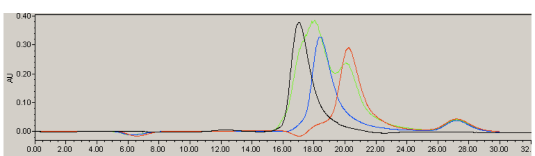
SYNZIP6:SYNZIP21 “off”

Alignment:

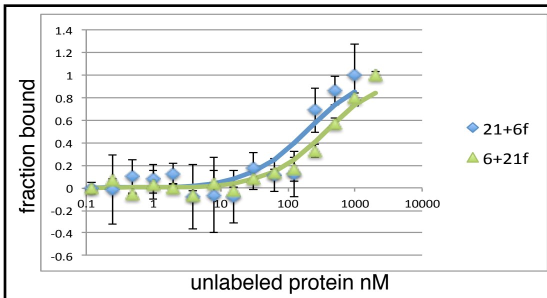
```
heptad position      fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abc
SZ6                  QK VAQLKNR VAYKLKE NAKLENI VARLEND NANLEKD IANLEKD IANLERD VAR
SZ21                 NE VAQLEND VAVIENE NAYLEKE IARLRKE IAALRDR LAHKK
hypothetical
```

Interaction Data

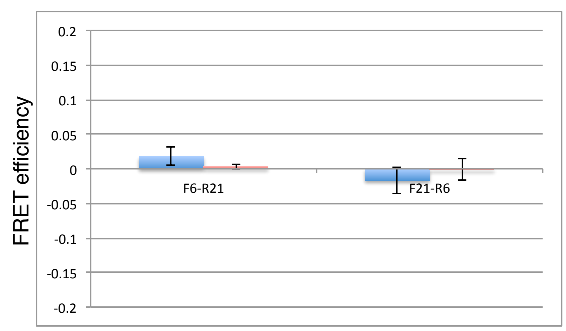
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
1/1	-	-	nd	multiple species	na	> 200 nM
				21 homodimer		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ21, red-SZ6



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ6: 4(pa,y2h), 5(pa,y2h),14 (y2h), 17(pa,y2h), 19(pa,y2h), 20(pa,y2h)

SZ21: 4(pa,y2h), 5(pa,y2h), 8(y2h), 10(pa,y2h), 11(pa,y2h), 12(y2h), 13(pa,y2h), 14(pa), 15(pa), 16(pa,y2h), 17(pa,y2h), 19(pa,y2h), 20(pa,y2h)

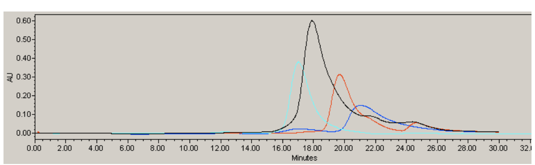
Additional notes

SYNZIP11:SYNZIP19

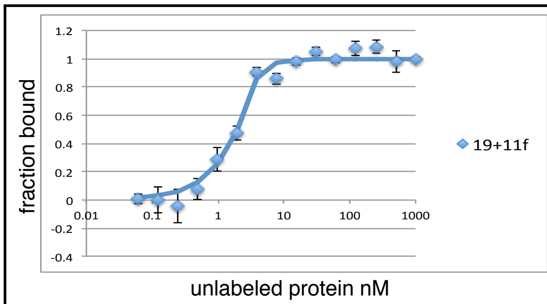
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg a
SZ11                EL TDELKNK KEALRKD NAALLNE LASLENE IANLEKE IAYFK
SZ19                NE LESLENK KEELKNR NEELKQK REQLKQK LAALRNK LDAYKNR L
hypothetical
```

Interaction Data

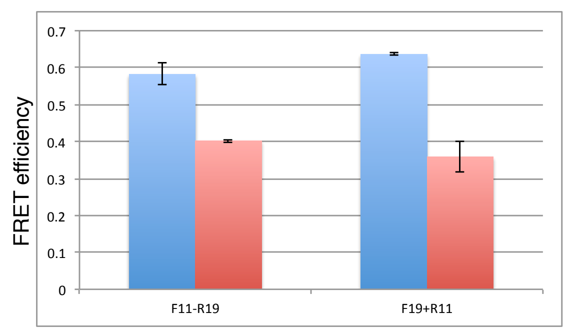
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.462/0.289	-	+++	nd	dimer	parallel	< 10 nM
	11 autoactivation	11 autoactivation		19 monomer potentially interact.w/ column		



Oligomerization(SEC): cyan- trimer control, black-mixture, blue-SZ19, red-SZ11



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ11:

SZ19: 2 (pa,y2h), 6 (pa,y2h), 12 (pa,y2h), 16 (pa,y2h), 18 (pa,y2h), 21 (pa,y2h), 22 (pa,y2h)

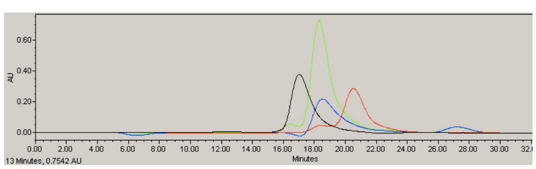
Additional notes

SYNZIP11:SYNZIP20

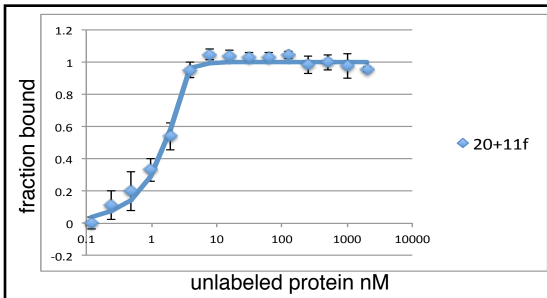
```
Alignment:
heptad position      fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg
SZ11                 EL TDELKNK KEALRKD NAALLNE LASLENE IANLEKE IAYFK
SZ20                 ST VEELLRA IQELEKR NAELKNR KEELKNL VAHLRQE LAAHKYE
hypothetical
```

Interaction Data

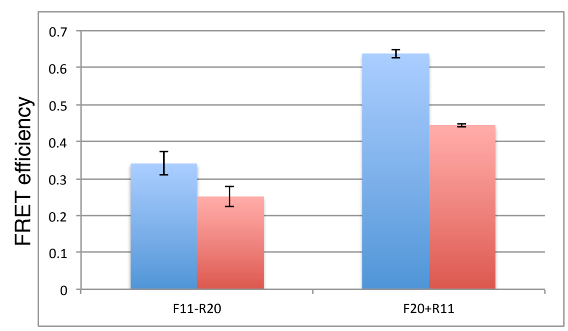
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0/0.062	+++	+++	nd	dimer	parallel	< 10 nM
	11 autoactivation	11 autoactivation		20 homodimer		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ20, red-SZ11



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ11: 1 (pa,y2h), 5 (pa,y2h), 17 (pa,y2h), 19 (pa,y2h), 21 (pa,y2h)

SZ20: 2 (pa,y2h), 3 (pa), 6 (pa,y2h), 12 (pa,y2h), 16 (pa,y2h), 18 (pa,y2h), 21 (pa,y2h), 22 (pa,y2h)

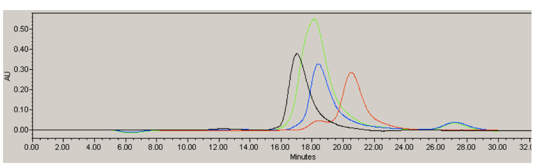
Additional notes

SYNZIP11:SYNZIP21

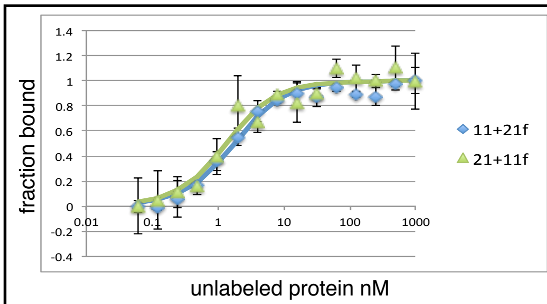
```
Alignment:
heptad position      fg abcdefg abcdefg abcdefg abcdefg abcdefg abcde
SZ11                 EL TDELKNK KEALRKD NAALLNE LASLENE IANLEKE IAYFK
SZ21                 NE VAQLEND VAVIENE NAYLEKE IARLRKE IAALRDR LAHKK
hypothetical
```

Interaction Data

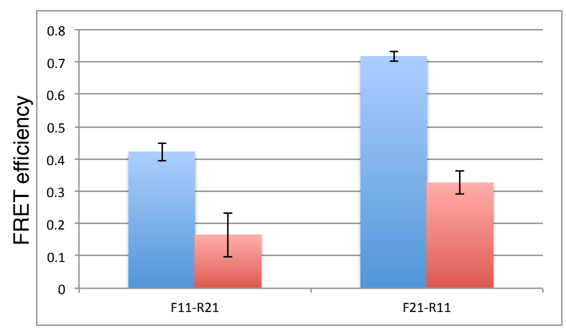
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.343/0.233	+	+++	nd	dimer	parallel	< 10 nM
	11 autoactivation	11 autoactivation		21 homodimer, runs as large dimer		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ21, red-SZ11



Affinity(FP): "f" denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ11: 1(pa,y2h), 5(pa,y2h), 17(pa,y2h), 19(pa,y2h), 20(pa,y2h)

SZ21: 4(pa,y2h), 5(pa,y2h), 8(y2h), 10(pa,y2h), 12(y2h), 13(pa,y2h), 14(pa), 15(pa), 16(pa,y2h), 17(pa,y2h), 19(pa,y2h), 20(pa,y2h)

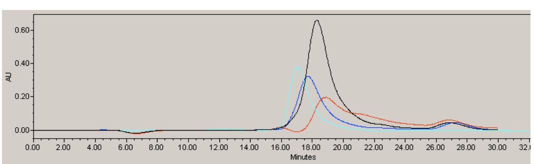
Additional notes

SYNZIP14:SYNZIP17

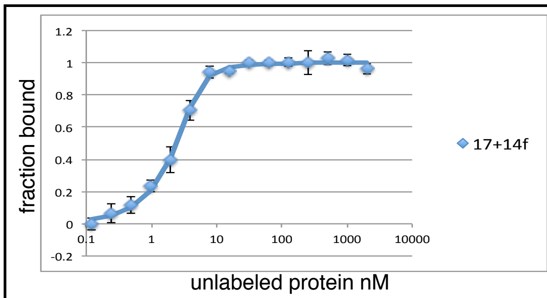
```
Alignment:
heptad position      fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abc
SZ14                 ND LDAYERE AEKLEKK NEVLRNR LAALENE LATLRQE VASMKQE LQS
SZ17                 NE KEELKSK KAELRNR IEQLKQK REQLKQK IANLRKE IEAYK
hypothetical
```

Interaction Data

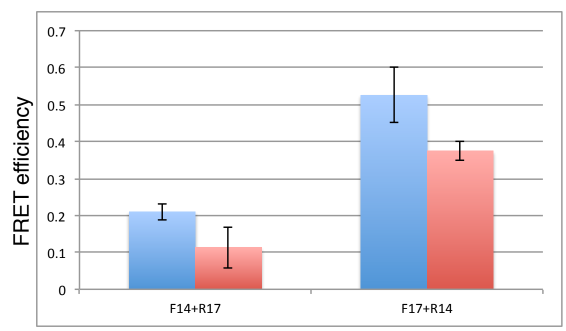
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.625/0.241	-	-	nd	dimer	parallel	< 10 nM
				14 homodimer, 17 interacts with column		



Oligomerization(SEC): cyan- trimer control, black-mixture, blue-SZ14, red-SZ17



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ14: 2(pa,y2h), 8(y2h), 12(pa,y2h), 15(pa), 21(pa), 22(pa,y2h)

SZ17: 3(pa,y2h), 6(pa,y2h), 11(pa,y2h), 12(pa,y2h), 15(pa), 18(pa,y2h), 21(pa,y2h), 22(pa,y2h)

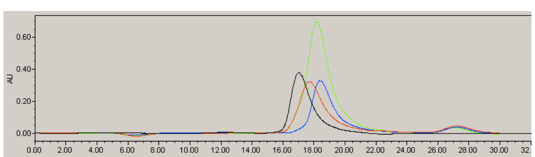
Additional notes

SYNZIP14:SYNZIP21

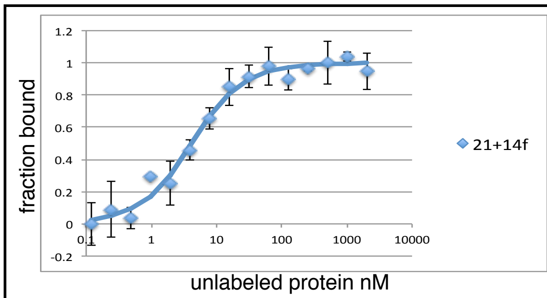
```
Alignment:
heptad position      fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abc
SZ14                 ND LDAYERE AEKLEKK NEVLRNR LAALENE LATLRQE VASMKQE LQS
SZ21                 NE VAQLEND VAVIENE NAYLEKE IARLRKE IAALRDR LAHKK
hypothetical
```

Interaction Data

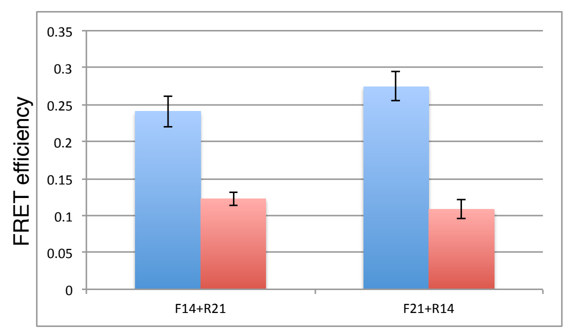
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.515/0.358	-	+	nd	dimer	parallel	< 10 nM
		only one direction		14 &21 homodimers		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ14, red-SZ21



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ14: 2 (pa,y2h), 8 (y2h), 12 (pa,y2h), 15 (pa), 17 (pa), 22 (pa,y2h)

SZ21: 4 (pa,y2h), 5 (pa,y2h), 8 (y2h), 10 (pa,y2h), 11 (pa,y2h), 12 (y2h), 13 (pa,y2h), 15 (pa), 16 (pa,y2h), 17 (pa,y2h), 19 (pa,y2h), 20 (pa,y2h)

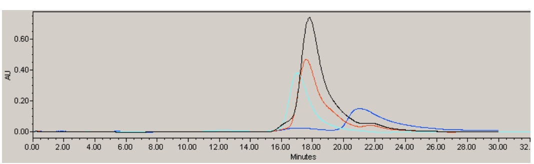
Additional notes

SYNZIP16:SYNZIP19

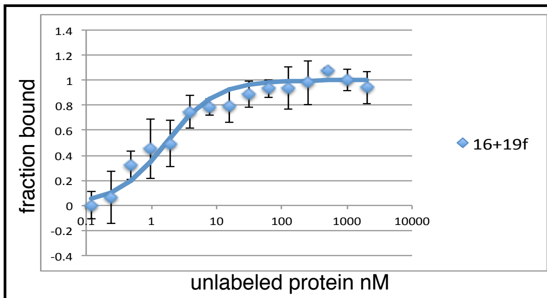
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg a
SZ16                NI LASLENK KEELKKL NAHLLKE IENLEKE IANLEKE IAYFK
SZ19                NE LESLENK KEELKNR NEELKQK REQLKQK LAALRNK LDAYKNR L
hypothetical
```

Interaction Data

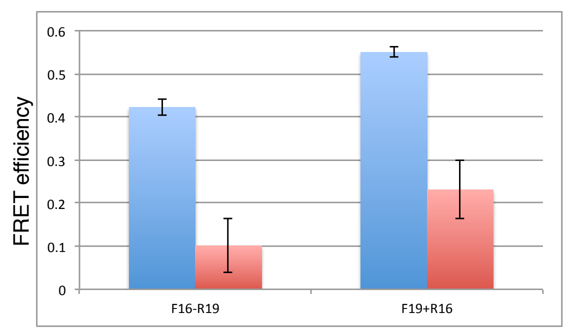
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.129/0.487	-	+++	nd	dimer	parallel	< 10 nM
		weak one direction		19 monomer potentially interact.w/ column, 16 homodimer		



Oligomerization(SEC): cyan- trimer control, black-mixture, blue-SZ19, red-SZ16



Affinity(FP): "f" denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ16: 5(pa,y2h), 20(pa,y2h), 21(pa,y2h)
SZ19: 2(pa,y2h), 6(pa,y2h), 11(pa,y2h), 12(pa,y2h), 18(pa,y2h), 21(pa,y2h), 22(pa,y2h)

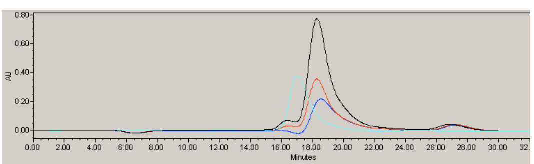
Additional notes

SYNZIP16:SYNZIP20

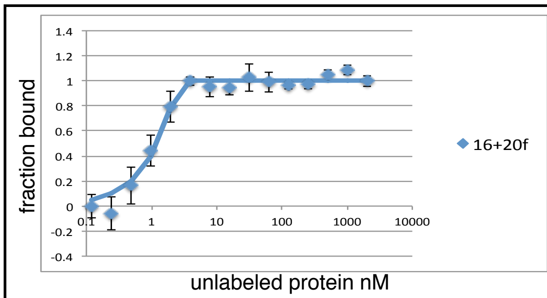
```
Alignment:
heptad position      fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg
SZ16                 NI LASLENK KEELKKL NAHLLKE IENLEKE IANLEKE IAYFK
SZ20                 ST VEELLRA IQELEKR NAELKNR KEELKNL VAHLRQE LAAHKYE
hypothetical
```

Interaction Data

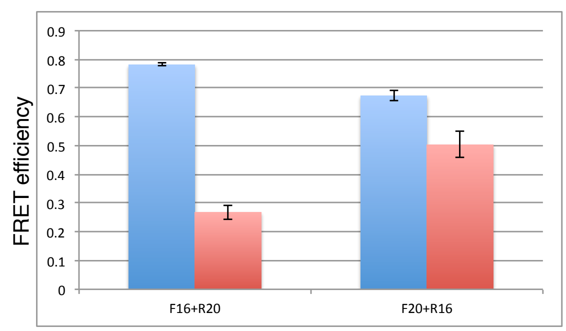
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.12/0.047	+++	+++	nd	dimer	parallel	< 10 nM
				16 & 20 homodimers		



Oligomerization(SEC): cyan- trimer control, black-mixture, blue-SZ16, red-SZ20



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ16: 5(pa,y2h), 19(pa,y2h), 21(pa,y2h)

SZ20: 2(pa,y2h), 3(pa), 6(pa,y2h), 11(pa,y2h), 12(pa,y2h), 18(pa,y2h), 21(pa,y2h), 22(pa,y2h)

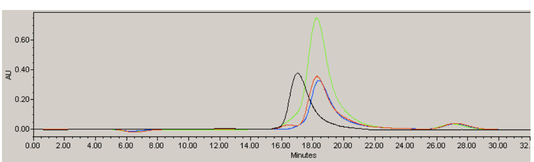
Additional notes

SYNZIP16:SYNZIP21

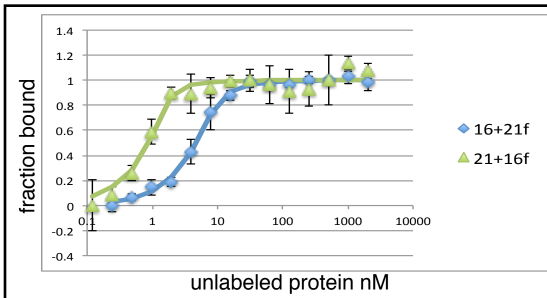
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdef
SZ16                NI LASLENK KEELKKL NAHLLKE IENLEKE IANLEKE IAYFK
SZ21                NE VAQLEND VAVIENE NAYLEKE IARLRKE IAALRDR LAHKK
hypothetical
```

Interaction Data

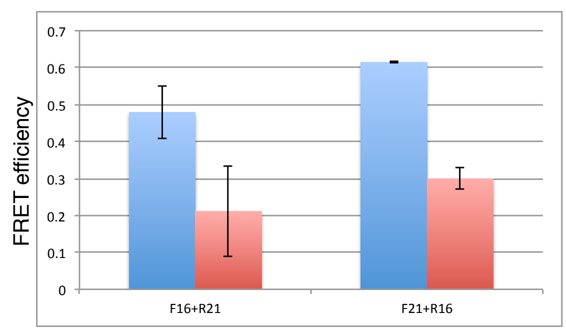
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.394/0.341	+++	+++	nd	dimer	parallel	< 10 nM
	no growth 1 direction			16 & 21 homodimers		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ16, red-SZ21



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ16: 5(pa,y2h), 19(pa,y2h), 20(pa,y2h)

SZ21: 4(pa,y2h), 5(pa,y2h), 8(y2h), 10(pa,y2h), 11(pa,y2h), 12(y2h), 13(pa,y2h), 14(pa), 15(pa), 17(pa,y2h), 19(pa,y2h), 20(pa,y2h)

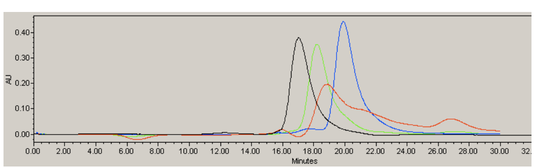
Additional notes

SYNZIP17:SYNZIP18

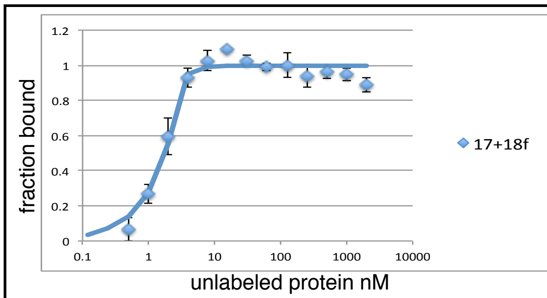
```
Alignment:
heptad position     fgabcdefgabcdefgabcdefgabcdefgabcdefgabcde
SZ17                NEKEELKSKKAELRNRIEQLKQKREQLKQKIANLRKEIEAYK
SZ18reverse           FYAEERELKALDRELNAIDKELRANENELRALDNELTAAIS
heptad position       dcbagfedcbagfedcbagfedcbagfedcbagfedcbagf
hypothetical
```

Interaction Data

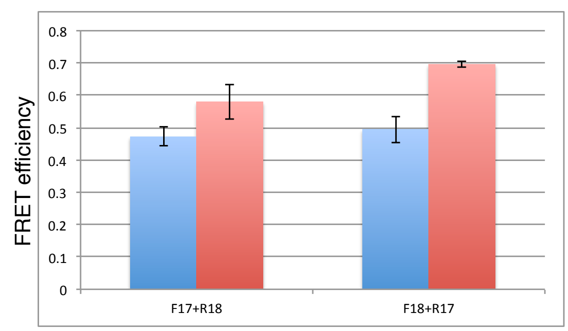
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.035/0.19	+++	+++	nd	dimer	antiparallel	< 10 nM
	18 autoactivation	18 autoactivation		17 interacts with column		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ18, red-SZ17



Affinity(FP): "f" denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ17: 3 (pa,y2h), 6 (pa,y2h), 11 (pa,y2h), 12 (pa,y2h), 14 (pa,y2h), 15 (pa), 21 (pa,y2h), 22 (pa,y2h)

SZ18: 4 (pa,*y2h*), 5 (pa,y2h), 13 (pa,y2h), 19 (pa,y2h), 20 (pa,y2h)

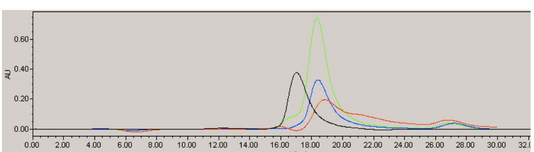
Additional notes

SYNZIP17:SYNZIP21

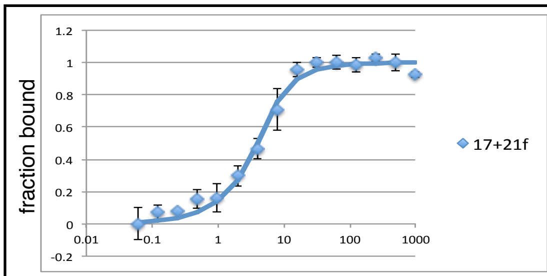
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcde
SZ17                NE KEELKSK KAELRNR IEQLKQK REQLKQK IANLRKE IEAYK
SZ21                NE VAQLEND VAVIENE NAYLEKE IARLRKE IAALRDR LAHKK
hypothetical
```

Interaction Data

Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.244/0.188	+++	+++	nd	dimer	parallel	< 10 nM
	weak 1 direction			21 homodimer, 17 interacts with column		

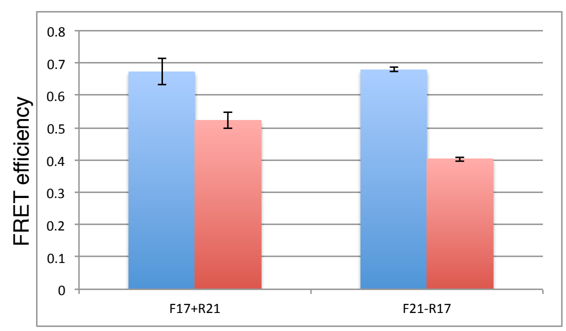


Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ21, red-SZ17



unlabeled protein nM

Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ17: 3(pa,y2h), 6(pa,y2h), 11(pa,y2h), 12(pa,y2h), 14(pa,y2h), 15(pa), 18(pa,y2h), 22(pa,y2h)
SZ21: 4(pa,y2h), 5(pa,y2h), 8(y2h), 10(pa,y2h), 11(pa,y2h), 12(y2h), 13(pa,y2h), 14(pa), 15(pa), 16(pa,y2h), 19(pa,y2h), 20(pa,y2h)

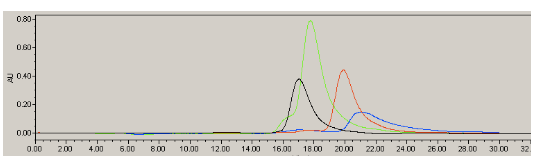
Additional notes

SYNZIP18:SYNZIP19

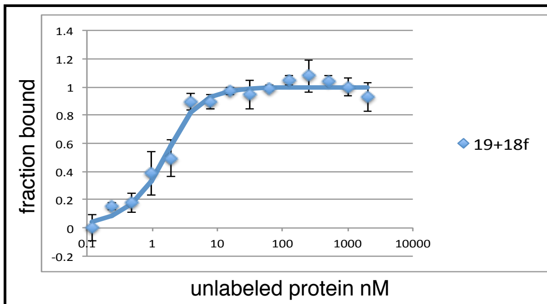
```
Alignment:
heptad position      fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg a
SZ18                 SI AATLEND LARLENE NARLEKD IANLERD LAKLERE EAYF
SZ19                 NE LESLENK KEELKNR NEELKQK REQLKQK LAALRNK LDAYKNR L
hypothetical
```

Interaction Data

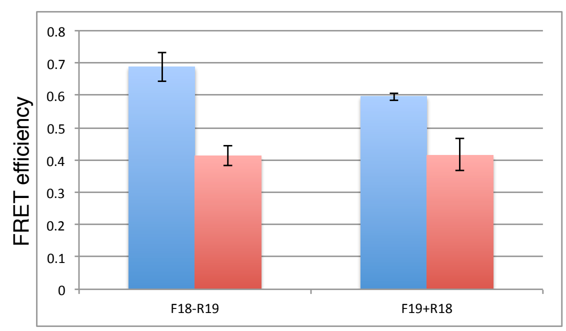
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0/0.107	+++	+++	0.376/0.263	dimer	parallel	< 10 nM
	18 autoactivation	18 autoactivation		19 monomer potentially interact.w/ column		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ19, red-SZ18



Affinity(FP): "f" denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ18: 4(pa,*y2h*), 5(pa,y2h), 13(pa,y2h), 17 (pa,y2h), 20(pa,y2h)

SZ19: 2(pa,y2h), 6(pa,y2h), 11(pa,y2h), 12 (pa,y2h), 16(pa,y2h), 21(pa,y2h), 22 (pa,y2h)

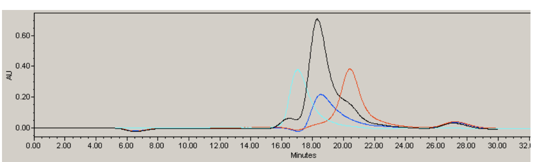
Additional notes

SYNZIP18:SYNZIP20

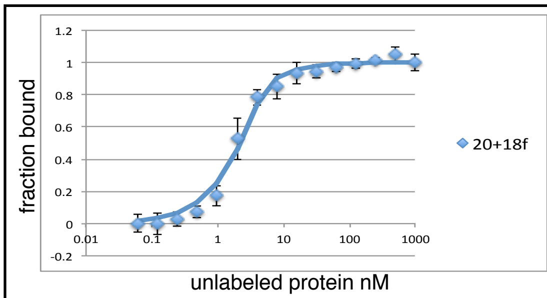
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg
SZ18                SI AATLEND LARLENE NARLEKD IANLERD LAKLERE EAYF
SZ20                ST VEELLRA IQELEKR NAELKNR KEELKNL VAHLRQE LAAHKYE
hypothetical
```

Interaction Data

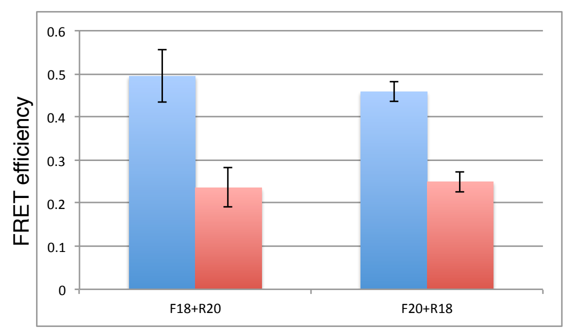
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.016/0.288	+++	+++	nd	dimer	parallel	< 15 nM
	18 autoactivation	18 autoactivation		20 homodimer		



Oligomerization(SEC): cyan- trimer control, greeblackn-mixture, blue-SZ20, red-SZ18



Affinity(FP): "f" denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ18: 4 (pa,*y2h*), 5 (pa,y2h), 13 (pa,y2h), 17 (pa,y2h), 19 (pa,y2h)

SZ20: 2 (pa,y2h), 3 (pa), 6 (pa,y2h), 11 (pa,y2h), 12 (pa,y2h), 16 (pa,y2h), 21 (pa,y2h), 22 (pa,y2h)

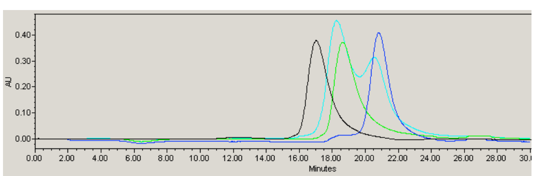
Additional notes

SYNZIP18: SYNZIP 21 “off”

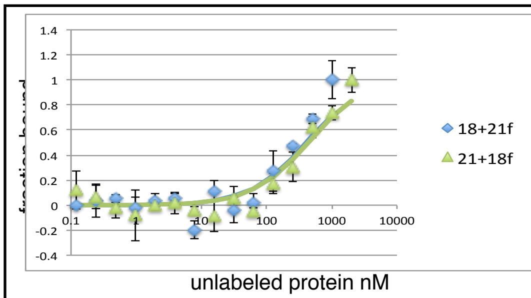
```
Alignment:
heptad position      fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg
SZ18                 SI AATLEND LARLENE NARLEKD IANLERD LAKLERE EAYF
SZ21                 NE VAQLEND VAVIENE NAYLEKE IARLRKE IAALRDR LAHKK
hypothetical
```

Interaction Data

Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
1.126/1.348	-	-	nd	no interaction	nd	>400 nM
	18 autoactivation	18 autoactivation		20 homodimer		



Oligomerization(SEC): black- trimer control, cyan-mixture, green-SZ20, blue-SZ18



Affinity(FP): “f” denotes labeled species

Potential interactors

SZ18: 4(pa,*y2h*), 5(pa,y2h), 13(pa,y2h), 17 (pa,y2h), 19(pa,y2h)

SZ21: 4(pa,y2h), 5(pa,y2h), 8(y2h), 10(pa,y2h), 11(pa,y2h), 12(y2h), 13(pa,y2h), 14(pa), 15(pa), 16(pa,y2h), 17(pa,y2h), 19(pa,y2h)

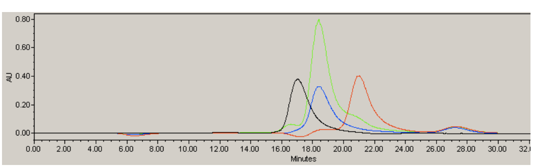
Additional notes

SYNZIP19:SYNZIP21

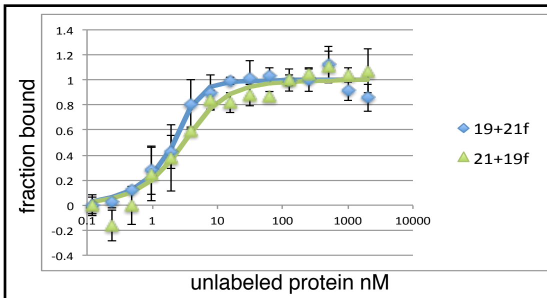
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg a
SZ19                NE LESLENK KEELKNR NEELKQK REQLKQK LAALRNK LDAYKNR L
SZ21                NE VAQLEND VAVIENE NAYLEKE IARLRKE IAALRDR LAHKK
hypothetical
```

Interaction Data

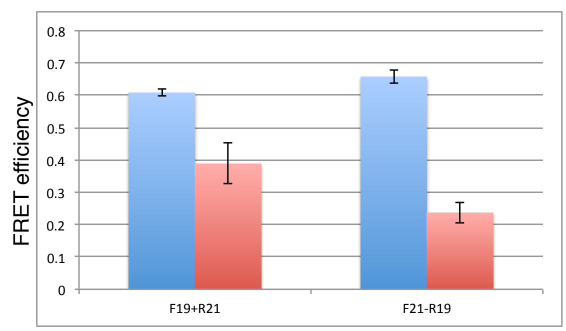
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.089/0	+++	+++	0.357/0.293	dimer	parallel	< 10 nM
				19 monomer potentially interact.w/ column, 21 homodimer		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ21, red-SZ19



Affinity(FP): "f" denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ19: 2(pa,y2h), 6(pa,y2h), 11(pa,y2h), 12(pa,y2h), 16(pa,y2h), 18(pa,y2h), 22(pa,y2h)

SZ21: 4(pa,y2h), 5(pa,y2h), 8(y2h), 10(pa,y2h), 11(pa,y2h), 12(y2h), 13(pa,y2h), 14(pa), 15(pa), 16(pa,y2h), 17(pa,y2h), 20(pa,y2h)

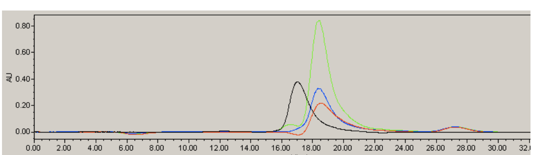
Additional notes

SYNZIP20:SYNZIP21

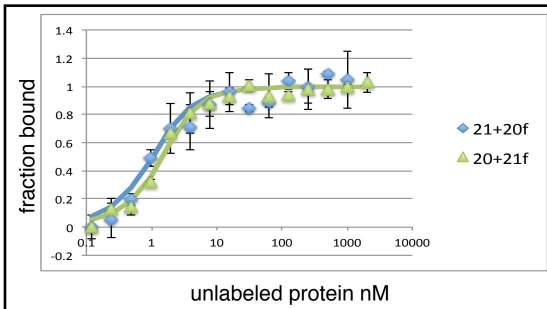
```
Alignment:
heptad position     fg abcdefg abcdefg abcdefg abcdefg abcdefg abcdefg abcdef
SZ20                ST VEELLRA IQELEKR NAELKNR KEELKNL VAHLRQE LAAHKYE
SZ21                NE VAQLEND VAVIENE NAYLEKE IARLRKE IAALRDR LAHKK
hypothetical
```

Interaction Data

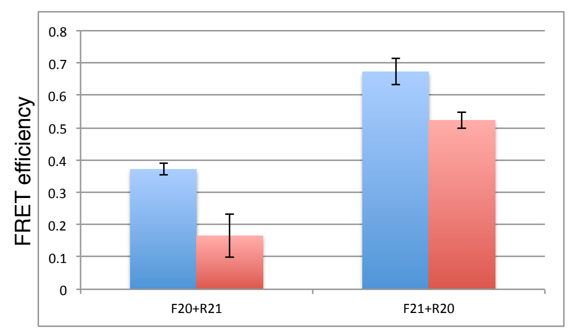
Protein microarray *arrayscore*	Y2H -Ura	Y2H -His	MAPK (fractional GFP intensity)	SEC	FRET	FP (K_d)
0.227/0.173	+++	+++	nd	dimer	parallel	< 10 nM
	slight growth variation			20 & 21 homodimers		



Oligomerization(SEC): black- trimer control, green-mixture, blue-SZ21, red-SZ20



Affinity(FP): “f” denotes labeled species



Orientation(FRET): blue- N-donor:N-acceptor, red- C-donor:N-acceptor

Potential interactors

SZ20: 2 (pa,y2h), 3 (pa), 6 (pa,y2h), 11 (pa,y2h), 12 (pa,y2h), 16 (pa,y2h), 18 (pa,y2h), 22 (pa,y2h)

SZ21: 4 (pa,y2h), 5 (pa,y2h), 8 (y2h), 10 (pa,y2h), 11 (pa,y2h), 12 (y2h), 13 (pa,y2h), 14 (pa), 15 (pa), 16 (pa,y2h), 17 (pa,y2h), 19 (pa,y2h)

Additional notes